

Math 196 Lecture Notes: Week 6

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1 Dot Products and Orthogonality in Higher Dimensions

1.1 The Dot Product and Geometry in the Plane

In this section of the course, we will discuss orthogonal projection in higher dimensions, and other associated phenomena. As a warm up, we start by reviewing the geometric significance of the dot product in the plane $\mathbb{R}^2$.

1.1.1 Length of vectors

Using the Pythagorean theorem, we can compute the length of a vector $|v|$. If

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

then  $|v|^2 = x^2 + y^2 = v \cdot v$, and so $|v| = \sqrt{v \cdot v}$.

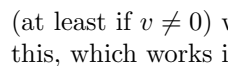
1.1.2 Orthogonality

Previously, we said that vectors v and w are at a right angle if $v \cdot w = 0$. One way to see this is that if

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

then the vectors w with $v \cdot w = 0$ are exactly the scalar multiples of

$$\begin{bmatrix} -y \\ x \end{bmatrix}$$

(at least if $v \neq 0$) which we know is v rotated by a right angle.  There is another way to see this, which works in higher dimensions as well. We first carry length into this setting.

1.2 Length and Angle in Higher Dimensions

Given a vector $v \in \mathbb{R}^n$, the Pythagorean theorem can be repeatedly applied to compute its length $|v|$: if

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

then

$$|v|^2 = x_1^2 + \cdots + x_n^2 = v \cdot v$$

Thus $|v| = \sqrt{v \cdot v}$, as in the plane. A useful property of length in general is that if $v \in \mathbb{R}^n$, and $a \in \mathbb{R}$, then

$$|av| = |a| \cdot |v|$$

This can be checked by direct computation using the formula above.

When are two vectors in $\mathbb{R}^n$ orthogonal? Well, this will happen when they form the legs of a right triangle. [Draw picture]. This means that

$$|v + w|^2 = |v|^2 + |w|^2$$

Expanding these in terms of dot products, we get

$$(v + w) \cdot (v + w) = v \cdot v + w \cdot w$$

The left side expands as

$$v \cdot v + 2(v \cdot w) + w \cdot w$$

so the vectors are orthogonal exactly when $v \cdot w = 0$, as is the case in $\mathbb{R}^2$.

1.2.1 Angle

Even if v and w are not orthogonal, their dot product still tells us about their angle with respect to each other. Consider the triangle with sides $v, w, v - w$. [Draw picture, with labeled angle θ] The *law of cosines* tells us that

$$|v - w|^2 = |v|^2 + |w|^2 - 2|v||w|\cos(\theta)$$

This is like the Pythagorean theorem with a correction for the angle. Expanding in terms of dot products, we get

$$v \cdot v - 2v \cdot w + w \cdot w = v \cdot v + w \cdot w - 2|v||w|\cos(\theta)$$

and so

$$v \cdot w = |v||w|\cos(\theta)$$

This gives us a geometric interpretation of the dot product. Vectors pointing in the same direction have dot product the product of their lengths, and vectors pointing in opposite directions have the negative of this, for example. If v and w are nonzero, then we can solve for θ :

$$\theta = \cos^{-1} \left(\frac{v \cdot w}{|v||w|} \right)$$

Example 1. What is the angle between $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$? We have $|u| = |v| = \sqrt{5}$, $u \cdot v = 1 \cdot 1 + 2 \cdot (-2) = -3$. Thus

$$\theta = \cos^{-1}(-3/5)$$

which is somewhere between $2\pi/3$ and $5\pi/6$.

Remark. Because $|\cos(\theta)| \leq 1$, we have the inequality

$$|v \cdot w| \leq |v||w|$$

Squaring both sides, we get

$$(v \cdot w)^2 \leq |v|^2 |w|^2$$

Expanding this in terms of the components of v and w , we have

$$\left[\sum_i x_i y_i \right]^2 \leq \left[\sum_i x_i^2 \right] \left[\sum_i y_i^2 \right]$$

This is called the *Cauchy-Schwarz inequality*. It is often useful to take $y_i = 1$ for all i , so the inequality becomes

$$\left[\sum_i x_i \right]^2 \leq n \sum_i x_i^2$$

1.2.2 Summary

For $v \in \mathbb{R}^n$, its length is computed as $|v| = \sqrt{v \cdot v}$. Two vectors v and w are orthogonal exactly when $v \cdot w = 0$. If v and w are both nonzero, then $|v||w| \neq 0$, and we can calculate the angle between v and w using

$$\cos(\theta) = \frac{v \cdot w}{|v||w|}$$

One could take these formulas as definitions of length, angle, and orthogonality in higher dimensions; this is the form in which we will work with these concepts. The preceding discussion was intended to motivate these definitions, by showing that they come from plane geometry.

With these concepts fixed, we can introduce some terminology that will be useful later: A *unit vector* is a vector u with $|u| = 1$. Given any nonzero vector v , we can *normalize* it to get a unit vector pointing in the same direction

$$u = \frac{1}{|v|} \cdot v$$

1.3 Transposes

Previously, we discussed how one could think about matrix multiplication in terms of dot products of rows and columns, in a somewhat informal way. We can now introduce an operation that makes this more formal, which is useful in general for thinking about dot products.

Definition. Given an $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

We define the *transpose* of A to be $n \times m$ matrix

$$A^\top = \begin{bmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & & \vdots \\ a_{1n} & \cdots & a_{mn} \end{bmatrix}$$

In other words, the columns of A become rows of $A^\top$ and vice versa.

Example 2.

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}^\top &= \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \\ \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 2 \end{bmatrix}^\top &= \begin{bmatrix} 1 & 3 & 0 \\ 2 & -1 & 2 \end{bmatrix} \\ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}^\top &= [1 \quad 0 \quad -1] \end{aligned}$$

In general, if $v \in \mathbb{R}^n$ is viewed as an $n \times 1$ matrix, then $v^\top$ is a $1 \times n$ row, which we can view as a linear map $\mathbb{R}^n \rightarrow \mathbb{R}$. This map is given by taking the dot product with v :

$$v^\top w = v \cdot w$$

Recall the proj_v can be defined by

$$\text{proj}_v(w) = \frac{v \cdot w}{v \cdot v} w$$

. In terms of length and transpose, we can write this

$$\frac{1}{|v|^2} v v^\top w$$

and so the matrix for proj_v can be written

$$\text{proj}_v = \frac{1}{|v|^2} v v^\top$$

Example 3. Given

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

we have $|v|^2 = 5$, and so

$$\text{proj}_v = \frac{1}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{bmatrix}$$

Later on, we will see a higher dimension analogue of this formula.

2 Orthonormal Sets and Orthogonal Projection

2.1 Orthonormal Sets of Vectors

A collection of vectors $u_1, \dots, u_r \in \mathbb{R}^n$ is said to be *orthonormal* if

1. each u_i is a unit vector: $|u_i| = 1$.
2. for $i \neq j$, u_i is orthogonal to u_j : $u_i \cdot u_j = 0$.

We can summarize this by saying that $u_i \cdot u_j = 1$ if $i = j$, and 0 otherwise.

Example 4. The standard basis $e_1, \dots, e_n$ is orthonormal.

Example 5. The columns of a rotation matrix

$$\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

are orthonormal.

Proposition. *An orthonormal set is linearly independent.*

Proof. The proof of this uses a generally useful property of orthonormal sets, given a linear combination

$$x_1 u_1 + \dots + x_r u_r$$

we can extract the coefficient x_i by taking the dot product with u_i .

$$u_i \cdot (x_1 u_1 + \dots + x_r u_r) = x_1 (u_i \cdot u_1) + \dots + x_r (u_i \cdot u_r)$$

Because $u_1, \dots, u_r$ is orthonormal, all of these are zero except $x_i (u_i \cdot u_i) = x_i$. In particular, if we have a linear relation

$$0 = x_1 u_1 + \dots + x_r u_r$$

then dotting with each u_i , we see that $x_1 = x_2 = \dots = x_r = 0$. □

We say $u_1, \dots, u_r$ is a *orthonormal basis* of a subspace V if $\text{span}\{u_1, \dots, u_r\} = V$. In particular, if $u_1, \dots, u_n \in \mathbb{R}^n$ is an orthonormal set of vectors, it is an orthonormal basis of $\mathbb{R}^n$. Given any vector v , the coordinates of v in this basis are the numbers $x_i = v \cdot u_i$.

2.2 Orthogonal Projection onto a Subspace

Previously, we saw how to compute the orthogonal projection of a vector onto a line in the plane. This is like the shadow that the vector casts on the line. [Picture] We can consider a similar operation in higher dimension space: projection onto a linear subspace.

First of all, given a vector v and a subspace W , we say that v is *orthogonal to W* if $v \cdot w = 0$ for all $w \in W$.

Example 6. In $\mathbb{R}^3$, e_3 is orthogonal to the xy -plane:

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = 0$$

Proposition. If $w_1, \dots, w_r$ is a basis for W , then v is orthogonal to W if and only if v is orthogonal to each w_i .

Proof. Any $w \in W$ can be written $x_1 w_1 + \dots + x_r w_r$. If v is orthogonal to each w_i , then

$$v \cdot w = \sum_i x_i (v \cdot w_i) = 0$$

Thus v is orthogonal to any vector in W . □

Definition. Given a subspace $W \subset \mathbb{R}^n$, its *orthogonal complement* $W^\perp$ is the set of all vectors v orthogonal to W .

Example 7. The orthogonal complement of a line in $\mathbb{R}^3$ is a plane, the orthogonal complement of a plane is a line. [Draw pictures]

Proposition. $W^\perp$ is a linear subspace.

Proof. This will be on the homework. □

We will see that any vector $v \in \mathbb{R}^n$ can be decomposed in a unique way as $v = v_1 + v_2$, where $v_1 \in W$, and $v_2 \in W^\perp$. The function sending v to its component in W is proj_W , orthogonal projection onto W . We will also see that this is linear. [Draw a picture with W a plane]

2.2.1 Computing Orthogonal Projection Using an Orthonormal Basis

If we happen to have an orthonormal basis $u_1, \dots, u_r$ for W , then we can use this to compute $\text{proj}_W(v)$. Next, we will study a method of constructing orthonormal bases, which will allow us to see that our definition of $\text{proj}_W(v)$ makes sense, and allows us to compute it.

Consider the vector

$$v_1 = \sum_i (v \cdot u_i) u_i$$

This is a linear combination of elements of W , so is in W . Moreover, because $u_1, \dots, u_r$ is orthonormal, we have

$$v_1 \cdot u_i = v \cdot u_i$$

for each i . Thus, if we set $v_2 = v - v_1$, we have $v_2 \cdot u_i = 0$ for each i , so $v_2 \in W^\perp$. Thus $v = v_1 + v_2$ is a decomposition of v into an element of W and an element of $W^\perp$, and so we can say that $v_1 = \text{proj}_W(v)$ in this case.

Example 8. What is the orthogonal projection of

$$w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

onto the plane

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}?$$

Let v_1, v_2 be the vectors spanning the plane, and note that v_1 and v_2 are orthogonal, $v_1 \cdot v_2 = 1 + 1 - 2 = 0$. Thus if we normalize these vectors, we get an orthogonal basis for the plane. Define

$$u_1 = \frac{1}{|v_1|} v_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$u_2 = \frac{1}{|v_2|} v_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Now our projection is the vector $(w \cdot u_1)u_1 + (w \cdot u_2)u_2$. The coefficients are

$$w \cdot u_1 = \frac{1 + 2 - 2}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

$$w \cdot u_2 = \frac{1 + 2 + 1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

We conclude that the projection is

$$\frac{1}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + \frac{4}{3} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \\ -1 \end{bmatrix}$$

We can express this formula for projection in terms of the transpose operation. If

$$U = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_r \\ \vdots & & \vdots \end{bmatrix}$$

then for any v , we have

$$U^T v = \begin{bmatrix} v \cdot u_1 \\ \vdots \\ v \cdot u_r \end{bmatrix}$$

and therefore

$$UU^T v = \sum_i (v \cdot u_i) u_i$$

Thus, the matrix for projection onto $W = \text{span}\{u_1, \dots, u_r\}$ is UU^T , when $u_1, \dots, u_r$ is orthonormal.

In the above example,

$$U = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{3} \\ 2/\sqrt{6} & -1/\sqrt{3} \end{bmatrix}$$

and so the matrix for proj_W is

$$UU^T = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{3} \\ 2/\sqrt{6} & -1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & -1/\sqrt{3} \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3 The Graham-Schmidt Process

As we have seen previously, if we have an orthonormal basis for W a subspace, we can use it to compute proj_W . The *Graham-Schmidt Process* is a method to construct an orthonormal basis, starting with any basis. This will also show us that any subspace has an orthonormal basis, which means that our definition of orthogonal projection makes sense. In this process, we will use orthogonal projection onto smaller subspaces, so the method is recursive.

3.1 Graham-Schmidt in Dimension Two

To illustrate the idea of the Graham-Schmidt process, I will explain how it can be used to construct orthonormal bases of $\mathbb{R}^2$, this being the simplest case that illustrates all the steps of the method.

Suppose we are given a basis v_1, v_2 for $\mathbb{R}^2$. In applying the Graham-Schmidt process, we will replace the vectors v_i with new vectors u_i , one by one, so that the u_i are orthonormal.

First, we need to normalize v_1 to make it a unit vector: we set

$$u_1 = \frac{1}{|v_1|}v_1$$

Now using u_1 , we can decompose v_2 into two components: one parallel to u_1 and the other orthogonal to it. Specifically, the parallel component is

$$\text{proj}_{u_1}(v_2) = (v_2 \cdot u_1)u_1$$

and the orthogonal component is

$$v_2 - \text{proj}_{u_1}(v_2) = v_2 - (v_2 \cdot u_1)u_1$$

Let's temporarily call the orthogonal component w_2 . Because v_1 and v_2 were a basis, v_2 is not in the span of u_1 , so w_2 must be nonzero. Thus we can normalize it, letting

$$u_2 = \frac{1}{|w_2|}w_2$$

Now u_1 and u_2 are unit vectors, and $u_1 \cdot u_2 = 0$, so u_1, u_2 is an orthonormal basis.

Example 9. Suppose we start with the basis

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

We then set

$$u_1 = \frac{1}{|v_1|}v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Then

$$w_2 = v_2 - (v_2 \cdot u_1)u_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -3/2 \\ 3/2 \end{bmatrix}$$

Finally, we set

$$u_2 = \frac{1}{|w_2|}w_2 = \frac{\sqrt{2}}{3} \begin{bmatrix} -3/2 \\ 3/2 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

[Draw pictures at each step of the process]

3.2 The Gram-Schmidt Process in General

For the Gram-Schmidt process in general, we are given a basis $v_1, \dots, v_r$ for a subspace $V \subset \mathbb{R}^n$, and we would like to replace the v_i with vectors $u_1, \dots, u_n$ such that they are still a basis for the same subspace V , and such that they are orthonormal. The idea is to construct each u_i one at a time, using the previously constructed u_i to compute projections.

Here is a formal description of the algorithm:

- At step 1:, normalize v_1 to get $u_1 = \frac{1}{|v_1|}v_1$.
- At step $m > 1$, we have already constructed $u_1, \dots, u_{m-1}$. Our construction will have ensured that $\text{span}\{u_1, \dots, u_{m-1}\} = \text{span}\{v_1, \dots, v_{m-1}\}$. Call this subspace V_{m-1} . Because v_m is independent from these vectors, it does not lie in V_{m-1} . Using $u_1, \dots, u_{m-1}$, we can decompose v_m into a component in V_{m-1} :

$$\text{proj}_{V_{m-1}}(v_m) = \sum_{i=1}^{m-1} (v_m \cdot u_i) u_i$$

and a component orthogonal to V_{m-1} :

$$w_m = v_m - \text{proj}_{V_{m-1}}(v_m) = v_m - \sum_{i=1}^{m-1} (v_m \cdot u_i) u_i$$

- To complete step m , we normalize w_m , setting $u_m = \frac{1}{|w_m|}w_m$. This is possible as $w_m \neq 0$.
- Once we finish step $r = \dim(V)$, we have $u_1, \dots, u_r$ an orthonormal basis of V .

Example 10. Let's use this to compute an orthonormal basis for

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^4$$

[Label the vectors v_1, v_2, v_3] We first set

$$u_1 = \frac{1}{|v_1|}v_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

Now we set

$$w_2 = v_2 - (v_2 \cdot u_1)u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$

In this case, $|w_2| = 1$, so $u_2 = w_2$. Finally, we compute

$$w_3 = v_3 - (v_3 \cdot u_1)u_1 - (v_3 \cdot u_2)u_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} - 0 \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

Thus, our orthonormal basis is

$$\begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

4 Projection, Orthogonal Complement, and Transposes

Now that we have a way of computing the projection onto a subspace, we can take a step back and think about some of its properties as a linear transformation.

Proposition. Let $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the orthogonal projection map onto a subspace W .

1. $\text{im}(T) = W$, so $\text{rk}(T) = \dim(W)$.
2. $\ker(T) = W^\perp$.
3. $P(w) = w$ for $w \in W$.
4. $P^2 = P$.

To see why these are true, recall that the projection $P(v)$ is defined as follows: v can be uniquely written as a sum $v = v_1 + v_2$ where $v_1 \in W$, $v_2 \in W^\perp$, and we set $P(v) = v_1$. From this, (2) and (3) follow immediately, and (1) and (4) follow from (3).

The subspaces W and $W^\perp$ are related in a particular way. By the rank nullity theorem, $\dim(W) + \dim(W^\perp) = n$. Moreover, we can say that $W \cap W^\perp = \{0\}$, and that any vector $v \in \mathbb{R}^n$ can be uniquely written as a sum of an element of W and an element of $W^\perp$. If we have $w_1, \dots, w_r$ a basis for W , and $v_1, \dots, v_s$ a basis for $W^\perp$, then $w_1, \dots, w_r, v_1, \dots, v_s$ is a basis for $\mathbb{R}^n$. With respect to such a basis, P takes the form

$$\begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix}$$

If we express a subspace W as $\text{im}(A)$ for A a matrix, then $W^\perp$ is related to the transpose of A .

Proposition. $\text{im}(A)^\perp = \ker(A^\top)$

Remark. If A is an $m \times n$ matrix, then $A^\top$ is $n \times m$, so both $\text{im}(A)^\perp$ and $\ker(A^\top)$ are subspaces of $\mathbb{R}^m$, and this equation makes sense.

Proof. Let $v_1, \dots, v_n$ be the columns of A . Then $w \in \text{im}(A)^\perp$ if and only if $v_i \cdot w = 0$ for each i . We can write $v_i \cdot w = v_i^\top w$. The rows of $A^\top$ are exactly the $1 \times m$ matrices $v_i^\top$, so $v_i^\top w$ is the i th entry of Aw . Therefore, $v_i^\top w = 0$ for each i if and only if $Aw = 0$. $\square$

If we apply the transpose twice, we get back to where we started: $(A^\top)^\top = A$. If we plug in $A^\top$ to the above formula, we get an equivalent formula

$$\text{im}(A^\top)^\perp = \ker(A)$$

We can write both of these formulas in yet another equivalent way, using the following property of orthogonal complements:

Proposition. For any subspace $W \subset \mathbb{R}^n$, $(W^\perp)^\perp = W$.

Proof. Let $r = \dim(W)$. We have seen that $\dim(W^\perp) = n - r$. The same argument implies that $\dim((W^\perp)^\perp) = n - (n - r) = r$. If $x \in W$, then by definition, $x \cdot y = 0$ for any $y \in W^\perp$, so $x \in (W^\perp)^\perp$. Thus $W \subset (W^\perp)^\perp$, and both subspaces have dimension r , so they must be equal. $\square$

Using this, we can write

$$\begin{aligned}\text{im}(A) &= \ker(A^\top)^\perp \\ \text{im}(A^\top) &= \ker(A)^\perp\end{aligned}$$

Together, these formulas give us a way of computing a representation of $W^\perp$ as an image or kernel from a representation of W .

Example 11. Consider the subspace

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3$$

We wish to describe $W^\perp$. We know that W is a plane, so $W^\perp$ should be a line. We can write

$$W = \text{im} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

and so

$$W^\perp = \ker \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

We can bring this 2×3 matrix to reduced row echelon form as follows:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

We have free variable z , with $x = z$, $y = -z$, and so a general element of the kernel looks like

$$\begin{bmatrix} z \\ -z \\ z \end{bmatrix} = z \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Thus

$$W^\perp = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

4.1 Algebraic Properties of Transpose

The transpose operation has some algebraic properties that will be useful to us.

1. $(A + B)^\top = A^\top + B^\top$
2. $(cA)^\top = cA^\top$
3. $(A^\top)^\top = A$
4. $(AB)^\top = B^\top A^\top$

The first three of these are straightforward to check from the definition. One can think of (1) and (2) as saying that transpose is linear as an operation on matrices. The fourth one is a bit more interesting. One way to see it is to divide A into rows and B into columns:

$$A = \begin{bmatrix} \dots & r_1 & \dots \\ & \vdots & \\ \dots & r_m & \dots \end{bmatrix}$$

$$B = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

Then the i, j entry of AB is $r_i v_j$, which is the j, i entry of $(AB)^\top$. On the other hand

$$A^\top = \begin{bmatrix} \vdots & & \vdots \\ r_1^\top & \dots & r_m^\top \\ \vdots & & \vdots \end{bmatrix}$$

$$B^\top = \begin{bmatrix} \dots & v_1^\top & \dots \\ & \vdots & \\ \dots & v_n^\top & \dots \end{bmatrix}$$

and so $B^\top A^\top$ has j, i entry $v_j^\top r_i^\top = r_i v_j$.

In particular, for the projection matrix $P = UU^\top$, we have $P^\top = UU^\top = P$. This is to say that P is a *symmetric matrix*, its entries a_{ij} satisfy $a_{ij} = a_{ji}$.

5 Orthogonal Matrices

Suppose that $u_1, \dots, u_n$ is an orthonormal basis for $\mathbb{R}^n$. We can package these into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_n \\ \vdots & & \vdots \end{bmatrix}$$

Because we started with a basis, this matrix is invertible. We can say more in this case. We will first show that A preserves the dot product, in the sense that

$$(Av) \cdot (Aw) = v \cdot w$$

for any vectors $v, w \in \mathbb{R}^n$. To see this, let

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad w = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Then

$$v \cdot w = \sum_i x_i y_i$$

On the other hand, we have

$$Av = x_1 u_1 + \dots + x_n u_n$$

$$Aw = y_1 u_1 + \dots + y_n u_n$$

and so, by the distributive property for dot product

$$(Av) \cdot (Aw) = \sum_{i,j} x_i y_j (u_i \cdot u_j)$$

now $u_i \cdot u_j$ is 0 if $i \neq j$, and 1 if $i = j$, so this comes out to

$$\sum_i x_i y_i = v \cdot w$$

As a consequence of A preserving the dot product, A preserves length, in the sense that $|Av| = |v|$ for any v . Moreover, A preserves orthogonality, in the sense that if v and w are orthogonal, so are Av and Aw .

In fact, any matrix A which preserves length also preserves orthogonality, because it must take the right triangle with sides $v, w, v + w$ to a right triangle with sides $Av, Aw, Av + Aw$. Such a matrix must take the standard basis $e_1, \dots, e_n$ to another orthonormal basis $u_1, \dots, u_n$. These u_i are the columns of A so we have come full circle.

Conclusion. *The following are equivalent, for A an $n \times n$ matrix:*

1. *The columns of A are an orthonormal basis.*
2. *$|Av| = |v|$ for all v .*
3. *$Av \cdot Aw = v \cdot w$ for all v, w .*

Such a matrix is said to be *orthogonal*. In $\mathbb{R}^2$, we have seen that rotation matrices are orthogonal. Thinking geometrically, reflection across any line through the origin preserves the length of vectors, so these are orthogonal as well. It turns out that the 2×2 orthogonal matrices are exactly the rotations and reflections. In $\mathbb{R}^3$, rotation about any axis is length preserving, so its matrix is orthogonal. The same goes for reflection across any plane through the origin. Again, these are all the orthogonal 3×3 matrices.

Another useful thing about orthogonal matrices is that they turn out to be easy to invert:

Proposition. *If A is orthogonal, then $A^\top = A^{-1}$.*

Proof. For any matrix A with columns $v_1, \dots, v_n$, the i, j entry in $A^\top A$ is $v_i^\top v_j = v_i \cdot v_j$. If A is orthogonal, this is 0 for $i \neq j$ and 1 for $i = j$, so $A^\top A = I_n$. Multiplying on the right by A^{-1} , we get $A^\top = A^{-1}$. $\square$

In fact, the condition that $A^\top = A^{-1}$ is equivalent to the other properties characterizing orthogonal matrices.

Proposition. *If A and B are orthogonal, then AB and A^{-1} are orthogonal.*

Proof. We check that AB and A^{-1} are length preserving:

$$|ABv| = |A(Bv)| = |Bv| = |v|$$

first using the fact that A is length preserving, then that B is length preserving. Similarly,

$$|v| = |AA^{-1}v| = |A^{-1}v|$$

using the fact that A is length preserving. $\square$